Soluções Folha de Exercícios 7

Exercício 3.

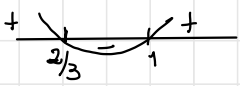
① $$\int_0^3 x^2\,dx = \left[\frac{x^3}{3}\right]_0^3 = \frac{3^3}{3} - \frac{0^3}{3} = 9$$

② $$\int_2^3 (3x^2 - 4x + 2)\,dx = \left[x^3 - 2x^2 + 2x\right]_2^3 = (27 - 18 + 6) - (8 - 8 + 4) = 11$$

③ $$\int_0^1 e^{\pi x}\,dx = \frac{1}{\pi}\int_0^1 \pi e^{\pi x}\,dx = \frac{1}{\pi}\left[e^{\pi x}\right]_0^1 =$$

$$= \frac{e^{\pi}}{\pi} - \frac{1}{\pi}$$

④ $\int_0^2 |(x-1)(3x-2)|\,dx = \int_0^{2/3} (x-1)(3x-2)\,dx + \int_{2/3}^1 -(x-1)(3x-2)\,dx + \int_1^2 (x-1)(3x-2)\,dx$



$= \int_0^{2/3} (3x^2-5x+2)\,dx + \int_{2/3}^1 (-3x^2+5x-2)\,dx + \int_1^2 (3x^2-5x+2)\,dx$

$= \left[x^3 - \frac{5x^2}{2} + 2x\right]_0^{2/3} + \left[-x^3 + \frac{5x^2}{2} - 2x\right]_{2/3}^1 + \left[x^3 - \frac{5x^2}{2} + 2x\right]_1^2$

$= \frac{55}{27}$.

⑤ Resolvido na aula.

⑥ Resolvido na aula

⑦ Resolvido na aula.

⑧ Resolvido na aula.

⑨ $\int_0^3 \sqrt{9-x^2}\,dx$, efetuando a substituição $x = 3\operatorname{sen} t$

(i) Substituição

Fazendo $x = 3\operatorname{sen} t$, tem-se

$\varphi(t) = 3\operatorname{sen} t$, $\varphi'(t) = 3\cos t$, $\varphi(0) = 0$, $\varphi(\pi/2) = 3$

(ii) Cálculo do novo integral

$$\int_0^3 \sqrt{9-x^2}\,dx = \int_0^{\pi/2} \sqrt{9-9\operatorname{sen}^2 t}\cdot \underbrace{3\cos t}_{\varphi'(t)}\,dt =$$

$$= \int_0^{\pi/2} \underbrace{3\sqrt{\cos^2 t}}_{|\cos t| = \cos t,\ t\in[0,\pi/2]} 3\cos t\,dt = \int_0^{\pi/2} 9\cos^2 t\,dt = 9\int_0^{\pi/2} \cos^2 t\,dt$$

$$= 9\int_0^{\pi/2} \frac{1+\cos(2t)}{2}\,dt = 9\int_0^{\pi/2} \frac{1}{2}\,dt + \frac{9}{2}\int_0^{\pi/2} \cos(2t)\,dt$$

$$= 9\int_0^{\pi/2} \frac{1}{2}\,dt + \frac{9}{4}\int_0^{\pi/2} 2\cos(2t)\,dt =$$

$$= 9\left[\frac{1}{2}t\right]_0^{\pi/2} + \frac{9}{4}\left[\operatorname{sen}(2t)\right]_0^{\pi/2} = \frac{9\pi}{4}$$

(16) $\int_{-5}^{0} 2x\sqrt{4-x}\,dx$, efetuando a substituição $t=\sqrt{4-x}$

(a) Substituição

Fazendo $t=\sqrt{4-x}$, tem-se $x=4-t^2$ e

$\varphi(t)=4-t^2$, $\varphi'(t)=-2t$, $\varphi(3)=-5$, $\varphi(2)=0$

(b) Cálculo do novo integral

$$\int_{-5}^{0} 2x\sqrt{4-x}\,dx = \int_{3}^{2} 2(4-t^2)\,t\cdot\underbrace{(-2t)}_{\varphi'(t)}\,dt =$$

$$= \int_{2}^{3} 2(4-t^2)\,t\,(2t)\,dt = \int_{2}^{3} (16t^2-4t^4)\,dt =$$

$$= \left[16\frac{t^3}{3}-4\frac{t^5}{5}\right]_2^3 = -\frac{1012}{15}$$

(11) $\int_0^2 x^3 e^{x^2} dx$

Sejam

$f'(x) = x e^{x^2}$ $\quad f(x) = \frac{1}{2} e^{x^2}$

$g(x) = x^2$ $\quad g'(x) = 2x$

Aplicando o método de integração por partes, tem-se

$$\int_0^2 x^3 e^{x^2} dx = \left[\frac{1}{2} x^2 e^{x^2}\right]_0^2 - \int_0^2 x e^{x^2} dx$$

$$= \left[\frac{1}{2} x^2 e^{x^2}\right]_0^2 - \frac{1}{2} \int_0^2 2x e^{x^2} dx$$

$$= \left[\frac{1}{2} x^2 e^{x^2}\right]_0^2 - \frac{1}{2} \left[e^{x^2}\right]_0^2$$

$$= \frac{3}{2} e^4 + \frac{1}{2}$$

(12) $\int_0^{\pi/2} e^{\operatorname{sen} x} \cdot \operatorname{sen} x \cos x \, dx$

Sejam

$f'(x) = e^{\operatorname{sen} x} \cos x$ $\quad f(x) = e^{\operatorname{sen} x}$

$g(x) = \operatorname{sen} x$ $\quad g'(x) = \cos x$

Aplicando o método de integração por partes, tem-se

$$\int_0^{\pi/2} e^{\operatorname{sen} x} \operatorname{sen} x \cos x \, dx = \left[e^{\operatorname{sen} x} \operatorname{sen} x\right]_0^{\pi/2} - \int_0^{\pi/2} e^{\operatorname{sen} x} \cdot \cos x \, dx$$

$$= \left[e^{\operatorname{sen} x} \cdot \operatorname{sen} x\right]_0^{\pi/2} - \left[e^{\operatorname{sen} x}\right]_0^{\pi/2} = e - (e-1) = 1$$

(13) Resolvido na aula

(14) $$\int_{-\pi}^{\pi} \cos^3 u \operatorname{sen} u \, du = -\int_{-\pi}^{\pi} \underbrace{\cos^3 u}_{f^3} \underbrace{(-\operatorname{sen} u)}_{f'} \, du =$$

$$= -\left[\frac{\cos^4 u}{4}\right]_{-\pi}^{\pi} = 0$$

(15) $$\int_{-4}^{0} t\sqrt{1+t^2}\, dt = \int_{-4}^{0} t\,(1+t^2)^{1/2}\, dt = \frac{1}{2}\int_{-4}^{0} 2t\,(1+t^2)^{1/2}\, dt$$

$$= \frac{1}{2}\left[\frac{(1+t^2)^{3/2}}{3/2}\right]_{-4}^{0} = \frac{1}{3} - \frac{1}{3}\,17^{3/2} = \frac{1}{3} - \frac{17}{3}\sqrt{17}$$

(16) $$\int_0^{\pi} x \operatorname{sen} x \, dx$$

Sejam

$f'(x) = \operatorname{sen} x$ $\quad f(x) = -\cos x$

$g(x) = x$ $\quad g'(x) = 1$

Aplicando a fórmula de integração por partes, temos que

$$\int_0^{\pi} x \operatorname{sen} x \, dx = \left[-x\cos x\right]_0^{\pi} - \int_0^{\pi} -\cos x \, dx$$

$$= \left[-x\cos x\right]_0^{\pi} + \left[\operatorname{sen} x\right]_0^{\pi} = \pi$$

(17) $\int_0^{\sqrt{2}/2} \operatorname{arcsen} x \, dx$

Sejam

$f'(x) = 1$ $\quad f(x) = x$

$g(x) = \operatorname{arcsen} x$ $\quad g'(x) = \frac{1}{\sqrt{1-x^2}}$

Aplicando o método de integração por partes, temos que

$$\int_0^{\sqrt{2}/2} \operatorname{arcsen} x \, dx = \left[x \operatorname{arcsen} x \right]_0^{\sqrt{2}/2} - \int_0^{\sqrt{2}/2} \frac{x}{\sqrt{1-x^2}} \, dx$$

$$= \left[x \operatorname{arcsen} x \right]_0^{\sqrt{2}/2} + \int_0^{\sqrt{2}/2} -x (1-x^2)^{-1/2} \, dx =$$

$$= \left[x \operatorname{arcsen} x \right]_0^{\sqrt{2}/2} + \frac{1}{2} \int_0^{\sqrt{2}/2} -2x (1-x^2)^{-1/2} \, dx$$

$$= \left[x \operatorname{arcsen} x \right]_0^{\sqrt{2}/2} + \frac{1}{2} \left[\frac{(1-x^2)^{1/2}}{1/2} \right]_0^{\sqrt{2}/2}$$

$$= \left[x \operatorname{arcsen} x \right]_0^{\sqrt{2}/2} + \left[\sqrt{1-x^2} \right]_0^{\sqrt{2}/2}$$

$$= \frac{\sqrt{2}}{2} \operatorname{arcsen} \frac{\sqrt{2}}{2} - 0 \cdot \operatorname{arcsen} 0 + \sqrt{1-1/2} - \sqrt{1}$$

$$= \frac{\sqrt{2}}{2} \cdot \frac{\pi}{4} + \sqrt{\frac{1}{2}} - 1 = \frac{\sqrt{2}\pi}{8} + \frac{\sqrt{2}}{2} - 1$$

(18) $$\int_{-\pi}^{\pi} \operatorname{sen}(2x)\cos x\,dx = \int_{-\pi}^{\pi} 2\operatorname{sen}x\cos^2 x\,dx =$$

$$= -2\int_{-\pi}^{\pi} (-\operatorname{sen}x)\cos^2 x\,dx = -2\left[\frac{\cos^3 x}{3}\right]_{-\pi}^{\pi} = 0$$

(19) $$\int_1^{e^3} \log t\,dt$$

Sejam $f(t) = 1$ $\quad f'(t) = t$

$g(t) = \log t$ $\quad g'(t) = \frac{1}{t}$

Aplicando o método de integração por partes, temos que

$$\int_1^{e^3} \log t\,dt = \left[t\log t\right]_1^{e^3} - \int_1^{e^3} 1\,dt =$$

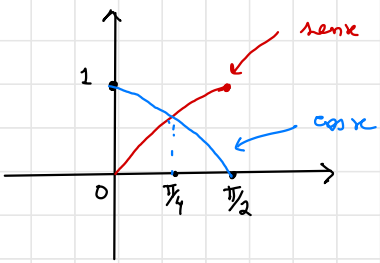
$$= \left[t\log t\right]_1^{e^3} - \left[t\right]_1^{e^3}$$

$$= e^3 \cdot 3 - (e^3 - 1) = 2e^3 + 1$$

(20) $$\int_0^{\pi/4} e^x\left(e^x + \frac{e^{-x}}{\cos^2 x}\right)dx = \int_0^{\pi/4}\left(e^{2x} + \frac{1}{\cos^2 x}\right)dx$$

$$= \left[\frac{1}{2}e^{2x} + \operatorname{tg}x\right]_0^{\pi/4} = \frac{1}{2}e^{\pi/2} + \frac{1}{2}$$

(21) $$\int_0^{\pi/2} |\operatorname{sen} x - \cos x|\,dx = \int_0^{\pi/4} -(\operatorname{sen} x - \cos x)\,dx + \int_{\pi/4}^{\pi/2} (\operatorname{sen} x - \cos x)\,dx$$



$$= \left[\cos x + \operatorname{sen} x\right]_0^{\pi/4} + \left[-\cos x - \operatorname{sen} x\right]_{\pi/4}^{\pi/2}$$

$$= 2(\sqrt{2} - 1)$$

(22) $$\int_{-3}^{2} \sqrt{|x|}\,dx = \int_{-3}^{0} \sqrt{-x}\,dx + \int_0^2 \sqrt{x}\,dx$$

$$= -\int_{-3}^{0} -(-x)^{1/2}\,dx + \int_0^2 x^{1/2}\,dx =$$

$$= -\left[\frac{(-x)^{3/2}}{3/2}\right]_{-3}^{0} + \left[\frac{x^{3/2}}{3/2}\right]_0^2 = \frac{2}{3}\left(2\sqrt{2} + 3\sqrt{3}\right)$$

(23) $$\int_0^2 f(x)\,dx = \int_0^1 x^2\,dx + \int_1^2 (2 - x)\,dx = \left[\frac{x^3}{3}\right]_0^1 + \left[2x - \frac{x^2}{2}\right]_1^2$$

$$= \frac{5}{6}$$

(24) $$\int_0^1 g(x)\,dx = \int_0^{1/2} x\,dx + \int_{1/2}^1 (1 - x)\,dx = \left[\frac{x^2}{2}\right]_0^{1/2} + \left[x - \frac{x^2}{2}\right]_{1/2}^1$$

$$= \frac{1}{4}$$

Exercício 4

Seja $F : \mathbb{R} \to \mathbb{R}$ definida por

$$F(x) = \int_0^x f(t)\,dt = -\frac{1}{2} + x^2 + x\,\text{sen}(2x) + \frac{\cos(2x)}{2}$$

Como a função f é contínua, pelo Primeiro Teorema Fundamental do Cálculo, a função F é derivável e

$$F'(x) = f(x)$$

Como $F(x) = -\frac{1}{2} + x^2 + x\,\text{sen}(2x) + \frac{\cos(2x)}{2}$, então

$$F'(x) = 2x + \text{sen}(2x) + 2x\cos(2x) - \text{sen}(2x)$$

Obtemos assim que

$$= 2x + 2x\cos(2x)$$

$$f(x) = 2x + 2x\cos(2x)$$

Então,

$$f\left(\frac{\pi}{4}\right) = 2\frac{\pi}{4} + 2\frac{\pi}{4}\cos\left(2\,\frac{\pi}{4}\right) = \frac{\pi}{2}$$

Derivando agora a função f, obtemos que

$$f'(x) = 2 + 2\cos(2x) - 4x\,\text{sen}(2x)$$

Então,

$$f'\left(\pi/4\right) = 2 - 4\frac{\pi}{4}\,\text{sen}\left(\frac{\pi}{2}\right) = 2 - \pi.$$

Exercício 5. Resolvido na aula.

Exercício 6. Resolvido na aula.

Exercício 7.

a) Como a função f é contínua, pelo Primeiro Teorema Fundamental do Cálculo, a função g é derivável e

$$g'(x) = f(x) - (2 + 4x)$$

Derivando a função g' (observe que f é derivável), obtemos

$$g''(x) = f'(x) - 4$$

b) Temos que:

$$g(0) = \int_0^0 f(t)\,dt - 0 = 0$$

$$g'(0) = f(0) - 2 = 2 - 2 = 0$$

$g''(x) = f'(x) - 4 > 0$, $\forall x \neq 0$ porque $f'(x) > 4$, $\forall x \neq 0$ por hipótese

c) Como $g''(x) > 0$ para todo o $x \neq 0$, a função g' é estritamente crescente (Notar que g' é contínua). Como $g'(0) = 0$ resulta que $g'(x) < 0$ para $x < 0$ e $g'(x) > 0$ para $x > 0$. Logo g é estritamente decrescente em $]-\infty, 0[$ e estritamente crescente em $]0, +\infty[$. Isto prova que em $x = 0$ a função tem um mínimo absoluto de valor $g(0) = 0$. Por outras palavras, $g(x) > 0$ para todo $x \neq 0$. Esta desigualdade equivale a

$$\int_0^x f(t)\,dt > 2x + 2x^2 \quad \text{para todo o } x \neq 0$$

Exercício 8:

a) Temos que

$$F(x) = 0 \iff \int_1^{x^2} e^{t^2}\,dt = 0$$

Como a função $g(t) = e^{t^2} > 0$, $\forall t \in \mathbb{R}$, temos que

$$\int_a^b g(t) = 0 \text{ se e só se } b = a$$

Então, $F(x) = 0 \iff x^2 = 1 \iff x = \pm 1$

b) $\mathrm{Dom}(f) = \mathbb{R}$. Temos que

$$F(-x) = \int_1^{(-x)^2} e^{t^2}\,dt = \int_1^{x^2} e^{t^2}\,dt = F(x), \quad \forall x \in \mathbb{R}$$

Então, F é uma função par

c) A função $f : \mathbb{R} \longrightarrow \mathbb{R}$, $t \longmapsto e^{t^2}$ é contínua

Então, usando o Primeiro Teorema Fundamental do Cálculo, temos que a função

$$F(x) = \int_1^{x^2} e^{t^2} \, dt$$

é derivável e

$$F'(x) = e^{x^4} \cdot 2x \quad , \quad x \in \mathbb{R}$$

Temos que

$$F'(x) = 0 \iff x = 0$$

		0	
F'	$-$	0	$+$
F	$\searrow$		$\nearrow$

Assim, F é estritamente decrescente no intervalo $]-\infty, 0[$ e é estritamente crescente no intervalo $]0, +\infty[$

Exercício 9

a) Por exemplo, a função $f : [0,1] \to \mathbb{R}$ definida por

$$f(x) = \begin{cases} 1 & \text{se } x \in \mathbb{Q} \cap [0,1] \\ 0 & \text{se } x \in (\mathbb{R} \setminus \mathbb{Q}) \cap [0,1] \end{cases}$$

b) Não existe. É suficiente observar que:

f derivável em $[0,1]$ $\Rightarrow$ f contínua em $[0,1]$ $\Rightarrow$ f integrável em $[0,1]$

c) Não existe. É suficiente observar que:

f derivável em $[0,1]$ $\Rightarrow$ f contínua em $[0,1]$ $\Rightarrow$ f primitivável em $[0,1]$

d) Ver o Exemplo 4 nos slides "Primitivas"

e) Por exemplo, a função $f : [0,1] \to \mathbb{R}$ definida por

$$f(x) = \begin{cases} 1 & \text{se } x \in [0, 1/2] \\ 0 & \text{se } x \in]1/2, 1] \end{cases}$$

Justificação:

- f não é fremetneáreel porque, pelo Teorema de Darboux, não pode ser a derevada de nenhuema fuençáo (com efeito, a fuençáo está defineda neem enterrvalo e náo foseee a fropriedade do valor enteromédo; anem, pelo Teorema de Darboux, náo pode sex a derevada de nenhuema fuençáo)

- f é enteqráreel porque tem eem neímero fuineto de fonto de descontineedade

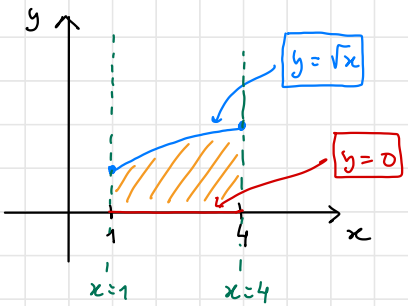
f) Tor exemflo, a fuençáo $f : [0,1] \longrightarrow \mathbb{R}$ defineda por

$$f(x) = \begin{cases} 1 & \text{se } x \in \mathbb{Q} \cap [0,1] \\ -1 & \text{se } x \in (\mathbb{R} \setminus \mathbb{Q}) \cap [0,1] \end{cases}$$

Obseneemos que $|f|(x) = 1, \ \forall x \in \mathbb{R}$.

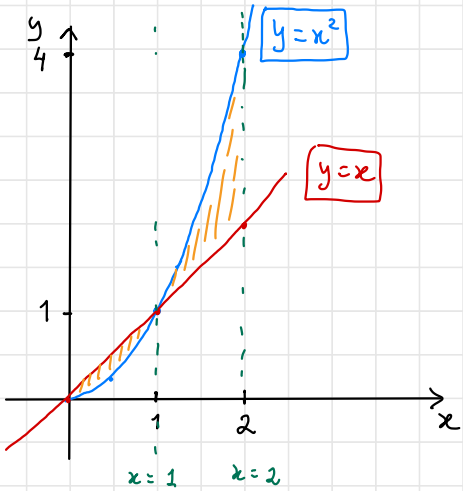
Exercício 10

a) $x = 1$, $x = 4$, $y = \sqrt{x}$, $y = 0$



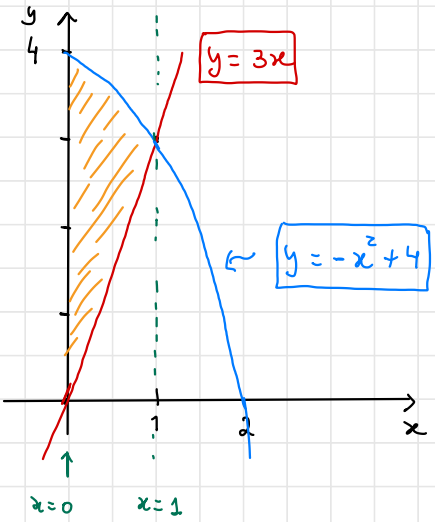
$$\text{Área} = \int_1^4 (\sqrt{x} - 0)\,dx = \int_1^4 x^{1/2}\,dx = \left[\frac{2}{3}x^{3/2}\right]_1^4 = \frac{14}{3}$$

b) $x = 0$, $x = 2$, $y = x$, $y = x^2$



$$\text{Área} = \int_0^1 (x - x^2)\,dx + \int_1^2 (x^2 - x)\,dx = 1$$

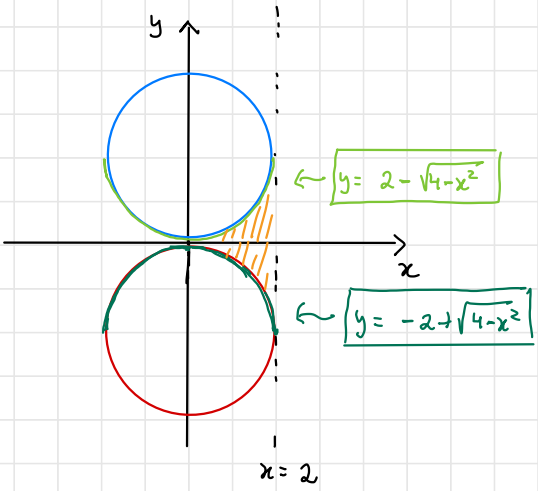
c) $x=0$, $x=1$, $y=3x$, $y=-x^2+4$



$$\text{Área} = \int_0^1 (-x^2+4-3x)\,dx = \left[-\frac{x^3}{3}+4x-\frac{3x^2}{2}\right]_0^1 = \frac{13}{6}$$

d) $x = 0$, $x = 2$, $x^2 + (y-2)^2 = 4$, $x^2 + (y+2)^2 = 4$

↳ $(0,2)$, $r = 2$ ↳ $(0,-2)$, $r = 2$



$x^2 + (y-2)^2 = 4 \Leftrightarrow (y-2)^2 = 4 - x^2$

$\Leftrightarrow y - 2 = \pm\sqrt{4-x^2}$

$\Leftrightarrow y = 2 \pm \sqrt{4-x^2}$

$x^2 + (y+2)^2 = 4 \Leftrightarrow (y+2)^2 = 4 - x^2$

$\Leftrightarrow y + 2 = \pm\sqrt{4-x^2}$

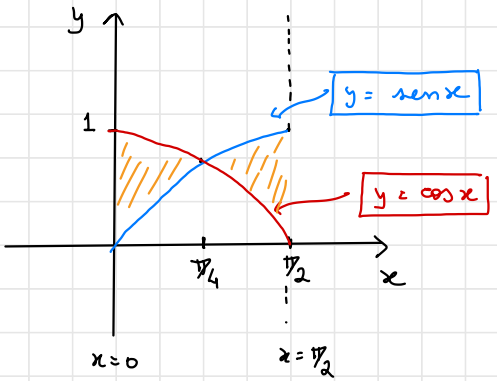
$\Leftrightarrow y = -2 \pm \sqrt{4-x^2}$

$$\text{Área} = \int_0^2 \left(\left(2 - \sqrt{4-x^2}\right) - \left(-2 + \sqrt{4-x^2}\right) \right) dx = 8 - 2\pi$$

Sugestão para calcular $\int_0^2 \sqrt{4-x^2}\, dx$: efetuar a substituição $x = 2\operatorname{sen} t$

(Ver o exercício 3 (9) que é análogo)

(e) $x = 0$, $x = \pi/2$, $y = \operatorname{sen} x$, $y = \cos x$



$$\text{Área} = \int_0^{\pi/4} (\cos x - \operatorname{sen} x)\,dx + \int_{\pi/4}^{\pi/2} (\operatorname{sen} x - \cos x)\,dx$$

$$= \left[\operatorname{sen} x + \cos x\right]_0^{\pi/4} + \left[-\cos x - \operatorname{sen} x\right]_{\pi/4}^{\pi/2} = 2\sqrt{2} - 2$$

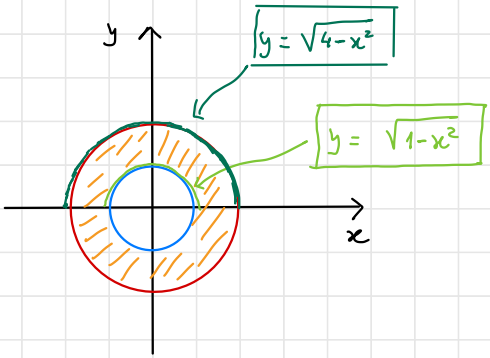
f) $x^2 + y^2 = 1$, $x^2 + y^2 = 4$

$x^2 + y^2 = 1 \iff y^2 = 1 - x^2$

$\iff y = \pm\sqrt{1 - x^2}$

$x^2 + y^2 = 4 \iff y^2 = 4 - x^2$

$\iff y = \pm\sqrt{4 - x^2}$

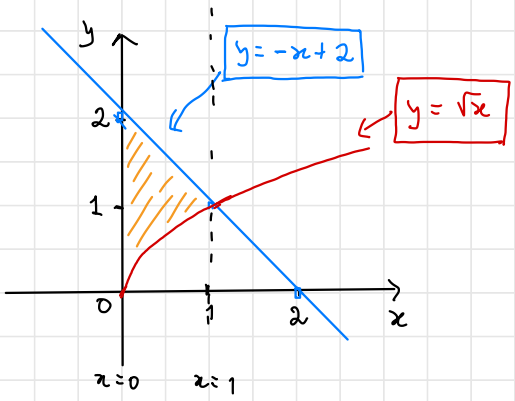


$$\text{Área} = 2\left(\int_{-2}^{2} \sqrt{4 - x^2}\,dx - \int_{-1}^{1} \sqrt{1 - x^2}\,dx\right) = 3\pi$$

Sugestão para calcular os integrais anteriores:

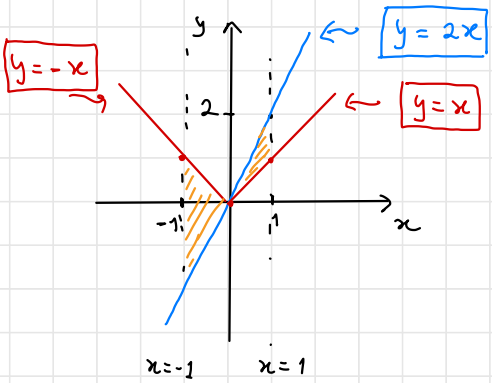
fazer as substituições $x = 2\operatorname{sen} t$ e $x = \operatorname{sen} t$, respetivamente

g) $x = 0$, $x = 1$, $y = \sqrt{x}$, $y = -x + 2$



$$\text{Área} = \int_0^1 (-x + 2 - \sqrt{x})\, dx = \frac{5}{6}$$

h) $x = -1$, $y = |x|$, $y = 2x$, $y = 1$

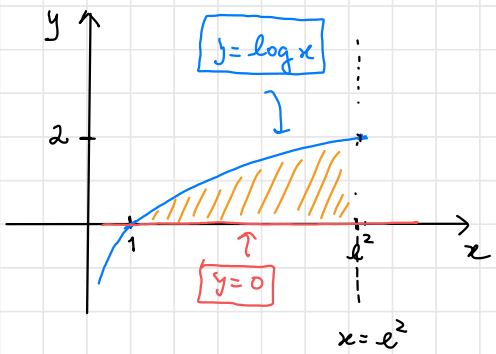


$$\text{Área} = \int_{-1}^{0} (-x - 2x)\,dx + \int_{0}^{1} (2x - x)\,dx =$$

$$= \int_{-1}^{0} -3x\,dx + \int_{0}^{1} x\,dx =$$

$$= \left[-\frac{3x^2}{2} \right]_{-1}^{0} + \left[\frac{x^2}{2} \right]_{0}^{1} = 2$$

i) $y = \log x$, $y = 0$, $x = e^2$



$$\text{Área} = \int_1^{e^2} (\log x - 0)\, dx = [x \log x]_1^{e^2} - \int_1^{e^2} 1\, dx$$

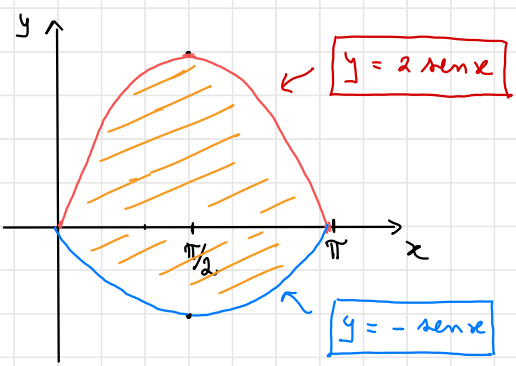
$f'(x) = 1$ $\quad f(x) = x$

$g(x) = \log x$ $\quad g'(x) = \frac{1}{x}$

$$= [x \log x]_1^{e^2} - [x]_1^{e^2}$$

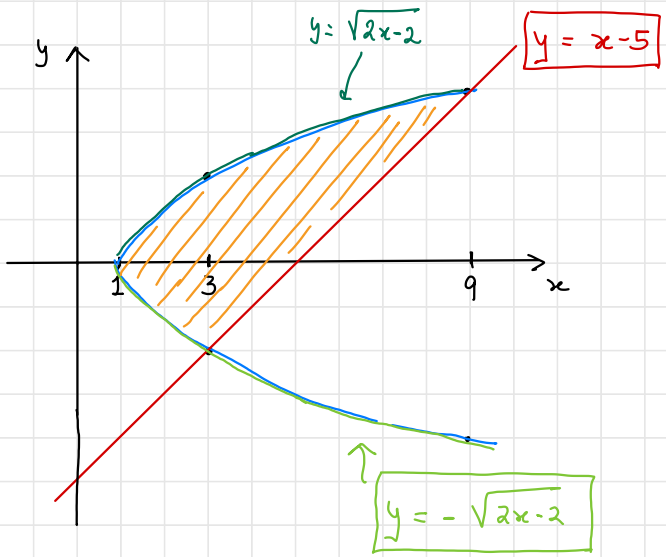
$$= e^2 + 1$$

g) $x = 0$, $x = \pi$, $y = 2 \operatorname{sen} x$, $y = -\operatorname{sen} x$



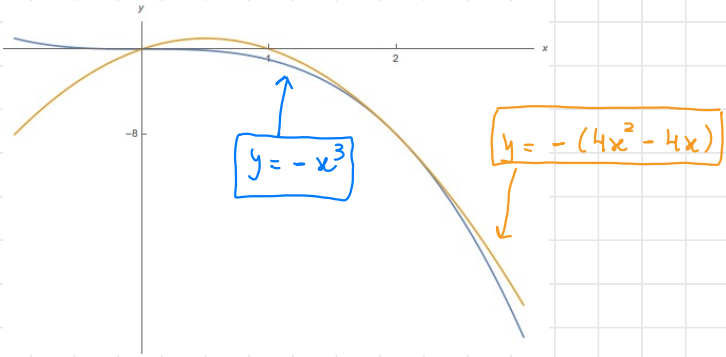
$$\text{Área} = \int_0^{\pi} (2 \operatorname{sen} x - 1 - \operatorname{sen} x)\, dx = \int_0^{\pi} 3 \operatorname{sen} x\, dx = [-3 \cos x]_0^{\pi} = 6$$

k) $y^2 = 2x - 2$, $y - x + 5 = 0$



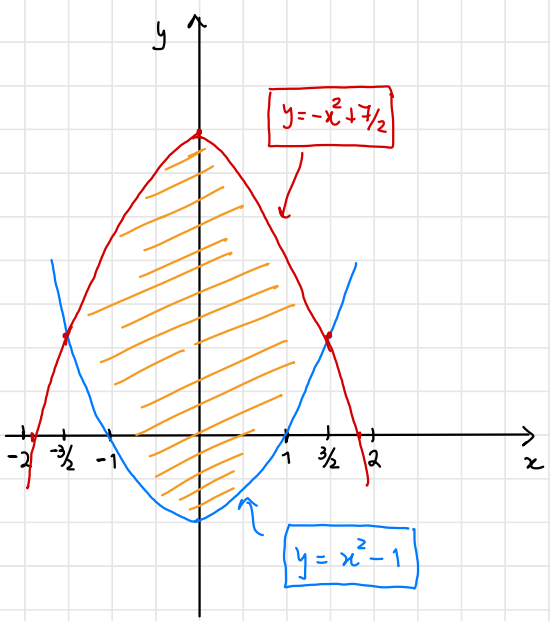
$$\text{Área} = \int_1^3 \left(\sqrt{2x-2} - \left(-\sqrt{2x-2}\right)\right) dx + \int_3^9 \left(\sqrt{2x-2} - (x-5)\right) dx = 18$$

l) $y = -x^3$, $y = -(4x^2 - 4x)$



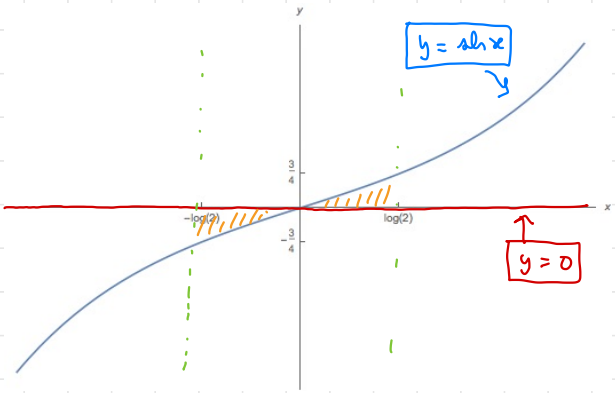
$$\text{Área} = \int_0^2 \left(-(4x^2 - 4x) - (-x^3)\right) dx = \frac{4}{3}$$

m) $y = -x^2 + \frac{7}{2}$, $y = x^2 - 1$



$$\text{Área} = \int_{-3/2}^{3/2} \left(-x^2 + \frac{7}{2} - (x^2 - 1)\right) dx = 9$$

m) $y = 0$, $x = -\log 2$, $x = \log 2$, $y = \operatorname{sh} x$



$$\text{Aire} = 2\int_0^{\log 2} (\operatorname{sh} x - 0)\,dx = 2\,[\operatorname{ch} x]_0^{\log 2}$$

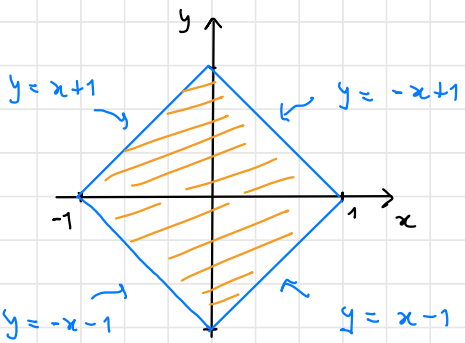
$$= 2\operatorname{ch}(\log 2) - 2\operatorname{ch} 0 = 2 \times \frac{5}{4} - 2 \times 1 = \frac{5}{2} - 2 = \frac{1}{2}$$

Exercício 11.

a) Resolvido na aula.

b) Resolvido na aula.

c) $D = \{(x,y) \in \mathbb{R}^2 : |x| + |y| \leq 1\}$



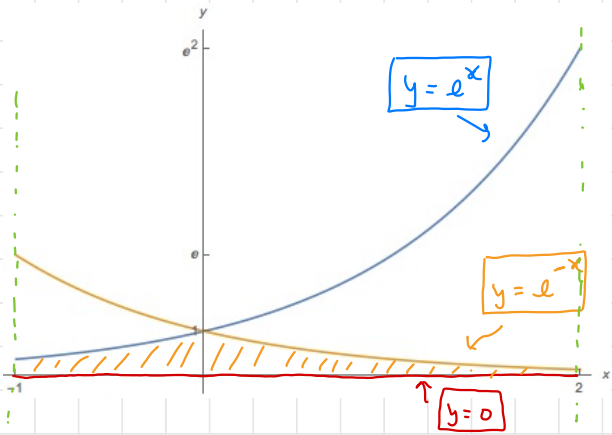
$$\text{Área}(D) = \int_{-1}^{0} (x+1-(-x-1))\,dx + \int_{0}^{1} (-x+1-(x-1))\,dx$$

ou

$$\text{Área}(D) = 4\int_{0}^{1} (-x+1-0)\,dx = 4\int_{0}^{1} (-x+1)\,dx$$

d) Resolvido na aula.

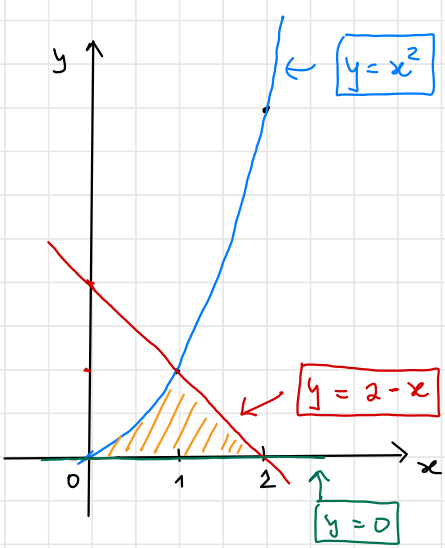
e) $D = \{(x,y) \in \mathbb{R}^2 \,.\, -1 \le x \le 2 \wedge 0 \le y \le e^x \wedge 0 \le y \le e^{-x}\}$



$$\text{Area}(D) = \int_{-1}^{0} (e^x - 0)\,dx + \int_{0}^{1} (e^{-x} - 0)\,dx =$$

$$= \int_{-1}^{0} e^x\,dx + \int_{0}^{1} e^{-x}\,dx$$

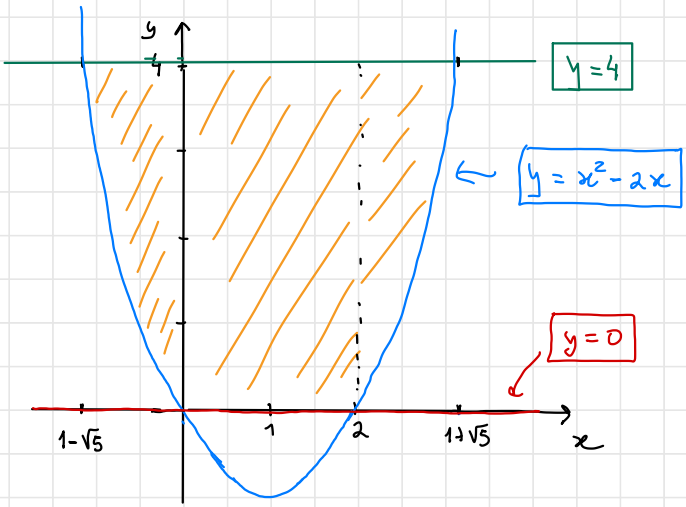
f) $D = \{(x,y) \in \mathbb{R}^2 \,.\, 0 \le x \le 2 \wedge 0 \le y \le x^2 \wedge 0 \le y \le 2 - x\}$



$$\text{Area}(D) = \int_{0}^{1} (x^2 - 0)\,dx + \int_{1}^{2} (2 - x - 0)\,dx$$

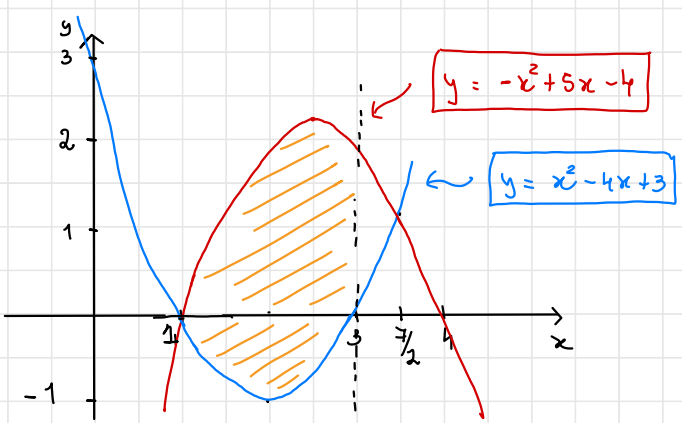
$$= \int_{0}^{1} x^2\,dx + \int_{1}^{2} (2 - x)\,dx$$

g) $D = \{(x,y) \in \mathbb{R}^2 : y \geq 0 \wedge y \geq x^2 - 2x \wedge y \leq 4\}$



$$\text{Area}(D) = \int_{1-\sqrt{5}}^{0} (4 - (x^2 - 2x))\,dx + \int_{0}^{2} 4\,dx + \int_{2}^{1+\sqrt{5}} (4 - (x^2 - 2x))\,dx$$

h) $D = \{(x,y) \in \mathbb{R}^2 : x \leq 3 \wedge x^2 - 4x + 3 \leq y \leq -x^2 + 5x - 4\}$



$$\text{Area}(D) = \int_{1}^{3} (-x^2 + 5x - 4) - (x^2 - 4x + 3)\,dx$$

Exercício 12 · Resolvido na aula.